Kreyszig Sec 1.3 Example 6:

Newton's Law of Cooling

$$\frac{dT}{dt} = -k(T - T_e)$$

T_e: temperature of environment (constant)

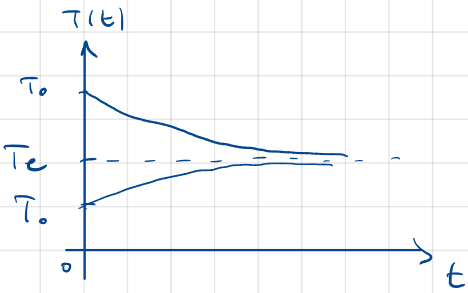
$$\int \frac{dT}{T - T_e} = -\int k\, dt$$

$$\ln(T - T_e) = -k \cdot t + C'$$

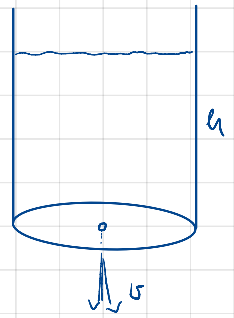
$$T = T_e + C \cdot e^{-kt}$$

$$T(0) = T_0 \Rightarrow T_e + C = T_0 \Rightarrow C = T_0 - T_e$$

$$T = T_e + (T_0 - T_e)e^{-kt} = T_0 + T_e(1 - e^{-kt})$$



Example 7: Leaking tank (Torricelli's Law)



$$\Delta V = A_{hole} \cdot v \cdot \Delta t = - A_{tank} \cdot \Delta h$$

$$v = \alpha \cdot \sqrt{2gh} \qquad \alpha \approx 0{,}6$$

$$\frac{\Delta h}{\Delta t} = - \frac{A_{hole}}{A_{tank}} \cdot \alpha \cdot \sqrt{2gh}$$

$\Delta t \to 0$:
$$h' = - \frac{A_h}{A_t} \cdot \alpha \sqrt{2g\, h} \qquad \text{ODE}$$

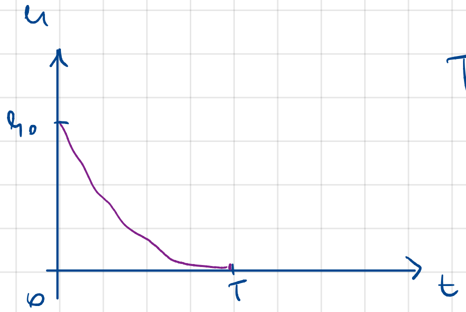
$$\int \frac{dh}{\sqrt{h}} = - \int \frac{A_h}{A_t} \alpha \sqrt{2g}\, dt$$

$$2\sqrt{h} = - A_h / A_t \; \alpha \sqrt{2g} \cdot t + C$$

$$h = \left(2C - 2A_h / A_t \; \alpha \sqrt{2g} \cdot t \right)^2$$

$t = 0$: $h = h_0 \Rightarrow h_0 = (2C)^2$

$$h(t) = \left(\sqrt{h_0} - 2\, A_h / A_t \; \alpha \sqrt{2g} \cdot t \right)^2 = 8g\, \alpha^2 \left(\frac{A_h}{A_t} \right)^2 \left(t - \sqrt{\frac{h_0}{8g}} \frac{A_t}{A_h} \frac{1}{\alpha} \right)^2$$



$$T = \sqrt{\frac{h_0}{8g}} \, \frac{A_t}{A_h} \cdot \frac{1}{\alpha}$$

Kreyszig Sec. 1.5 Problems

3.) $y' - y = 5.2$

Homogeneous eq. : $y' - y = 0 \Rightarrow y = C \cdot e^{x}$

a particular solution : $y = -5.2$

$\Big\}\ y = Ce^{x} - 5.2$

Recall : $y'(x) + p(x) \cdot y(x) = r(x)$

$$y(x) = e^{-h(x)}\left(\int e^{h(x)} r(x)\, dx + C\right), \text{ where } h = \int p(x)\, dx$$

Here $p(x) = -1$, $r(x) = 5.2$

$h(x) = \int (-1)\, dx = -x$

$$y(x) = e^{x}\left(\int e^{-x} \cdot 5.2\, dx + C\right) = C \cdot e^{x} - 5.2$$

4.) $y' = 2y - 4x$

Hom. eq. : $y' - 2y = 0 \Rightarrow y = C \cdot e^{2x}$

Part. sol. : $y = 2x + 1$

using the formula : $p = -2$, $r = -4x$

$h = -2x$

$$y = e^{2x}\left(\int e^{-2x} (-4x)\, dx + C\right) = \cancel{e^{2x}} \cdot \cancel{e^{-2x}} (2x+1) + Ce^{2x}$$

5) $y' + ky = e^{-kx}$

Hom.: $y' = -ky \Rightarrow y = C \cdot e^{-kx}$

part.: $y = x \cdot e^{-kx}$ $\quad / \ y' = e^{-kx} + x \cdot (-k) e^{-kx} = e^{-kx} - k \cdot y \ /$

formula: $p = k$, $r = e^{-kx}$ $\quad \rightarrow h = k \cdot x$

$$y = e^{-kx}\left(\int e^{kx} e^{-kx}\, dx + C\right) = e^{-kx} \cdot (x + C)$$

9.) $\quad y' + y\sin x = e^{\cos x}$

$p = \sin x \ , \quad r = e^{\cos x}$

$\hookrightarrow \quad h = \int \sin x \, dx = -\cos x$

$$y = e^{\cos x}\left(\int e^{-\cos x} \cdot e^{\cos x} \, dx + C \right) = e^{\cos x}(x + C)$$

Hom. solution:

$$y' + y \sin x = 0$$

$$\int \frac{dy}{y} = - \int \sin x \, dx$$

$$\ln y = \cos x + C' \quad \rightarrow \quad y = \overset{\overset{e^{C'}}{\shortparallel}}{C} \cdot e^{\cos x}$$

7. Second order ODE-s

Linear: $y'' + p(x)y' + q(x)y = r(x)$

Homogeneous: $r(x) \equiv 0$

$\hookrightarrow$ y_1 and y_1 are solutions $\longrightarrow$ $c_1 y_1 + c_2 y_2$ is also a solution

For general sol'n we need two linearly independent solutions:

$k_1 y_1(x) + k_2 y_2(x) = 0 \quad \forall x \in I \quad \Rightarrow \quad k_1 = k_2 = 0$

/ I: some interval /

c_1 and c_2 are fixed by 2 initial conditions

K 2.1 / 17

$$4x^2 y'' - 3y = 0 \qquad y(1) = -3 \qquad y'(1) = 0$$

$\rightarrow \quad y_1 = x^{3/2}, \qquad y_2 = x^{-1/2}$

Check:

$$4x^2 y_1'' - 3y_1 = 4x^2 \cdot 3/2 \cdot 1/2 \, x^{-1/2} - 3 \cdot x^{3/2} = 0 \checkmark$$

$$4x^2 y_2'' - 3y_2 = 4x^2 \cdot (-1/2)(-3/2) \cdot x^{-5/2} - 3 \cdot x^{-1/2} = 0 \checkmark$$

$$y(x) = c_1 \cdot y_1(x) + c_2 y_2(x)$$

$$y(1) = c_1 \cdot 1^{3/2} + c_2 \cdot 1^{-1/2} = c_1 + c_2 \overset{!}{=} 1$$

$$y'(1) = c_1 \cdot 3/2 \cdot 1^{1/2} + c_2 (-1/2) \cdot 1^{-3/2} = 3/2 \, c_1 - 1/2 \, c_2 \overset{!}{=} 0$$

$$c_2 = 1 - c_1 \quad \Rightarrow \quad 3/2 \, c_1 - 1/2 (1 - c_1) = 0$$

$$c_1 = 1/4 \quad \rightarrow \quad c_2 = 3/4$$

18.) $x^2y'' - xy' + y = 0$ $\quad y(1) = 4.3$, $y'(1) = 0{,}5$

$y_1 = x$, $y_2 = x \cdot \ln x$ $\longrightarrow$ $y_2' = \ln x + x \cdot 1/x = \ln x + 1$, $y_2'' = 1/x$

check:

$x^2 y_1'' - x y_1' + y_1 = x^2 \cdot 0 - x \cdot 1 + x = 0$ ✓

$x^2 y_2'' - x \cdot y_2' + y_2 = \cancel{x^2 \cdot 1/x} - x \cdot (\cancel{\ln x} + \cancel{1}) + \cancel{x \ln x} = 0$ ✓

$y(x) = C_1 \cdot y_1(x) + C_2 \cdot y_2(x)$

$y(1) = c_1 \cdot 1 + c_2 \cdot 1 \cdot \ln 1 = c_1 = 4.3$

$y'(1) = c_1 + c_2(\ln 1 + 1) = c_1 + c_2 = 0.5$

$c_1 = 4.3 \longrightarrow c_2 = -3.8$

3.) $y'' + y' = 0$

y doesn't appear! $\longrightarrow$ reduction of order : $u = y'$

$u' + u = 0$

$$\int \frac{du}{u} = -\int dx$$

$\ln u = -x + C_1$ (ln

$u = C_1 e^{-x}$

$y = \int u \, dx = -C_1 e^{-x} + C_2$

In general : Reduction of order

find a particular solution y_1 $\longrightarrow$ generate another (y_2) :

$y_2 = u \cdot y_1 \longrightarrow y_2' = u' y_1 + u y_1'$, $y_2'' = u'' y_1 + 2u' y_1' + u y_1''$

9.) $x^2 y'' - 5xy' + 9y = 0$

$y_1 = x^3$ / Check: $x^2 \cdot 3 \cdot 2 \cdot x - 5x \cdot 3x^2 + 9x^3 = 0$ ✓ /

$$y_2(x) = u(x)\, y_1(x) = u(x) \cdot x^3$$

$$y_2' = u' \cdot x^3 + u \cdot 3 \cdot x^2 \quad , \quad y_2'' = u'' x^3 + 2 \cdot u' \cdot 3x^2 + u \cdot 6x$$

$$\hookrightarrow \; x^2 (x^3 u'' + 6x^2 u' + 6x \cdot u) - 5x(x^3 u' + 3x^2 u) + 9x^3 u = 0$$

$$x^5 u'' + 6x^4 u' + \cancel{6x^3 u} - 5x^4 u' - \cancel{15 x^3 u} + \cancel{9x^3 u} = 0$$

Divide by x^5, $v = u'$:

$$v' + 1/x \cdot v = 0$$

$$\int \frac{dv}{v} = -\int \frac{dx}{x} \longrightarrow \ln v = -\ln x \longrightarrow v = 1/x$$

$$u = \int v \, dx = \ln x$$

$$y = c_1 \cdot x^3 + c_2 \cdot x^3 \cdot \ln x$$

Constant coefficients:

$$y'' + ay' + by = 0$$

$y = e^{\lambda x} \rightarrow$ characteristic eqn.:

$$\lambda^2 + a\lambda + b = 0$$

$$\lambda_{1,2} = \frac{-a \pm \sqrt{a^2 - 4b}}{2}$$

- $\lambda_1, \lambda_2 \in \mathbb{R}$ $\quad a^2 > 4b$: $\quad y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$
- $\lambda_1 = \lambda_2^* \notin \mathbb{R}$ $\quad a^2 < 4b$: $\quad y = c_1 e^{(-a/2 + i\omega)x} + c_2 e^{(-a/2 - i\omega)x}$

$$= e^{-\frac{ax}{2}} (c_1 + c_2) \cos(\omega x) + e^{-\frac{ax}{2}} (c_1 - c_2)\, i \sin(\omega x)$$

real if $c_1 = c_2^*$!

- $\lambda_1 = \lambda_2 \in \mathbb{R}$ $\quad a^2 = 4b$

K.22/23) $y'' + y' - 6y = 0 \qquad y(0) = 10 \qquad y'(0) = 0$

$a = 1, \; b = -6$

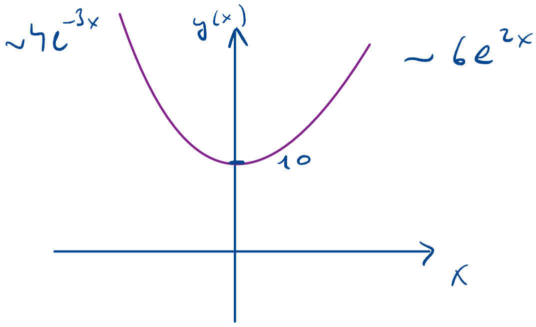
$$\lambda_{1,2} = \frac{-1 \pm \sqrt{1+4\cdot 6}}{2} = \frac{-1\pm 5}{2} \to 2, \; -3$$

$$y = c_1 \cdot e^{2x} + c_2 \cdot e^{-3x}$$

$$y(0) = c_1 + c_2 \overset{!}{=} 10$$

$$y'(0) = 2c_1 - 3c_2 \overset{!}{=} 0 \qquad 2c_1 - 3(10 - c_1) = 0$$

$$c_1 = 6 \qquad c_2 = 4$$



30.) $9y'' - 30y' + 25y = 0 \qquad y(0) = 3.3 \qquad y'(0) = 10$

$$y'' - \frac{30}{9}y' + \frac{25}{9}y = 0$$

$$a = -\frac{30}{9} \qquad b = \frac{25}{9}$$

$$\lambda_{1,2} = \frac{30/9 \pm \sqrt{30^2/9^2 - 4\cdot 1 \cdot 25/9}}{2} = \frac{5}{3}$$

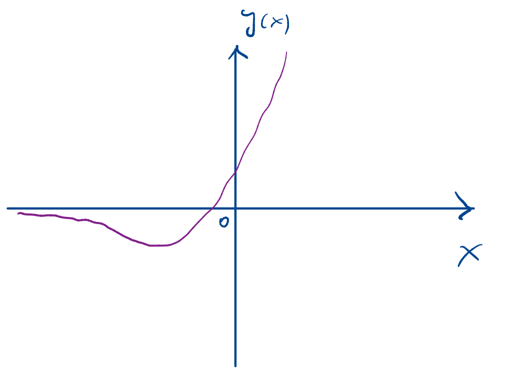
$$y = c_1 \cdot e^{5/3\,x} + c_2 \cdot x \cdot e^{5/3\,x}$$

$$y(0) = c_1 = 3.3$$

$$y'(0) = c_1 \, 5/3 \, e^{5/3\cdot 0} + c_2 \cdot e^{5/3 \cdot 0} + c_2 \cdot 0 \cdot 5/3 \cdot e^{5/3\cdot 0} = 5/3\, c_1 + c_2 = 10$$

$$\to c_2 = 4.5$$

$$y = 3.3 \cdot e^{5/3 x} + 4.5 \cdot x \cdot e^{5/3 \cdot x}$$



22.) $y'' + 4y' + (\pi^2+4)y = 0 \qquad y(1/2)=1, \qquad y'(1/2) = -2$

$a=4, \qquad b = \pi^2+4$

$$\lambda_{1,2} = \frac{1}{2}\left(-4 \pm \sqrt{4^2 - 4\cdot 1 \cdot (\pi^2+4)}\right) = -2 \pm i\pi$$

$$y = c_1 \cdot e^{(-2+i\pi)x} + c_2 \cdot e^{(-2-i\pi)x} \qquad (*)$$

$$y(1/2) = c_1 \cdot e^{(-2+i\pi)1/2} + c_2 \cdot e^{(-2-i\pi)\cdot 1/2} = i e^{-1}(c_1 - c_2) = 1$$

$$y'(1/2) = c_1(-2+i\pi)\cdot e^{-1}\cdot i + c_2(-2-i\pi)e^{-1}(-i) = -2$$

$$-2\underbrace{ie^{-1}(c_1-c_2)}_{1} + i\pi \cdot i e^{-1}(c_1+c_2) = -2$$

$$\rightarrow c_2 = -c_1$$

$$ie^{-1}2c_1 = 1 \Rightarrow c_1 = -i/2\, e$$

$$y = -i/2\, e \cdot e^{(-2+i\pi)x} + i/2\, e \; e^{(-2-i\pi)x}$$

$$= e \cdot e^{-2x} \frac{e^{i\pi x} - e^{-i\pi x}}{2i} = e^{1-2x} \cdot \sin(\pi x)$$

OR, more simply:

$$y = e^{-2x}\left(A \sin(\pi x) + B\cos(\pi x)\right)$$

$$y(1/2) = e^{-1}\cdot(A\cdot 1 + B\cdot 0) = e^{-1}\cdot A = 1 \Rightarrow A = e$$

$$y'(1/2) = -2\cdot e^{-1}(A\cdot 1 + B\cdot 0) + e^{-2}(A\cdot\pi\cdot 0 - B\cdot\pi\cdot 1) = -2$$

$$= -2\cdot e^{-1}\cdot \underset{e}{\underset{\shortparallel}{A}} + e^{-2}\cdot(-B) = -2$$

$$\hookrightarrow B = 0$$

Linear (in)dependence

Wronskian: $W = y_1 y_2' - y_2 y_1' = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$

$$x^2 y'' - 3xy' + 3y = 0 \qquad x \in [-1,1]$$

$$y_1 = \cos(3 \arccos x)$$
$$y_2 = 6x - 8x^3$$

Check: Home work :)

NOW: $y_1' = -\sin(3 \arccos x) \cdot 3 \cdot \frac{-1}{\sqrt{1-x^2}}$

$$y_2' = 6 - 24x^2$$

$$W = \cos(3 \arccos x) \cdot (6 - 24x^2) - (6x - 8x^3) \cdot 3 \frac{\sin(3 \arccos x)}{\sqrt{1-x^2}}$$

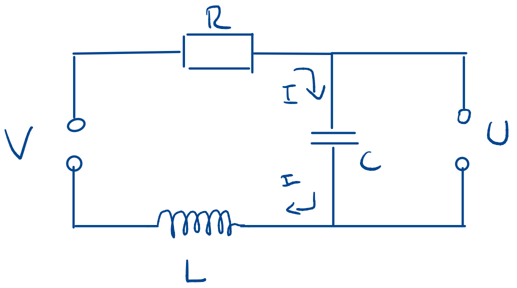
$x = 0$: $\cos(3 \cdot \pi/2) \cdot 6 - 0 = 0$

$x = 1/2$: $\cos(3 \cdot \pi/3) \cdot \underbrace{(6 - 24 \cdot 1/4)}_{0} - (3-1)\, 3 \frac{\overbrace{\sin(3 \cdot \pi/3)}^{0}}{\sqrt{3}/2} = 0$

So they are <u>not</u> linearly independent!

$$\cos(3 \arccos x) = 4x^3 - 3x \quad ! \qquad \text{Chebyshev polynomial}$$

RLC circuit



$$V = R\cdot I + L\cdot \dot I + Q/C$$
$$\qquad\qquad\qquad\quad \overset{\shortparallel}{U_C = U}$$

$$Q = C\cdot U \longrightarrow I = \dot Q = C\cdot \dot U$$

$$U = R\cdot C\cdot \dot U + L\cdot C\cdot \ddot U + U$$

initial state : $V = U_0 = \text{const.} \Rightarrow I = 0 \Rightarrow \dot U = 0$

$$U = U_0$$

So,
$$\ddot U + \frac{R}{L}\dot U + \frac{1}{L\cdot C} U = 0$$

initial cond.: $U(0) = U_0$, $\dot U(0) = 0$

$$\lambda_{1,2} = \frac{1}{2}\left(-R/L \pm \sqrt{(R/L)^2 - \frac{4}{LC}}\right) = -\frac{R}{2L}\left(1 \mp \sqrt{1 - \frac{4L}{R^2C}}\right)$$

- $\frac{R^2C}{4L} > 1$: overdamped $\lambda_1, \lambda_2 \in \mathbb{R}$

$$U(t) = c_1\cdot e^{\lambda_1 t} + c_2 e^{\lambda_2 t} \qquad \lambda_1, \lambda_2 < 0 \ (!)$$

$$U(0) = C_1 + C_2 = U_0$$

$$\dot U(0) = \lambda_1 c_1 + \lambda_2 c_2 = 0 \longrightarrow C_2 = -\frac{\lambda_1}{\lambda_2} C_1$$

$$\left(1 - \frac{\lambda_1}{\lambda_2}\right) C_1 = U_0$$

$$U(t) = \frac{\lambda_2}{\lambda_2 - \lambda_1} U_0\, e^{\lambda_1 t} - \frac{\lambda_1}{\lambda_2 - \lambda_1} U_0 \cdot e^{\lambda_2 t}$$

- $\frac{R^2C}{4L} > 1$: underdamped

$$U(t) = e^{-\alpha t}\left(A \cdot \cos(\omega t) + B \sin(\omega t)\right)$$

$$\alpha = \frac{R}{2L}$$

$$\omega = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

$$U(0) = A \overset{!}{=} U_0$$

$$\dot{U}(0) = -\alpha \cdot A + B\omega \overset{!}{=} 0 \quad \rightarrow B = \frac{\alpha}{\omega} U_0$$

$$U(t) = U_0\, e^{-\alpha t}\left(\cos(\omega t) + \alpha/\omega \sin(\omega t)\right)$$

- $\frac{R^2C}{4L} = 1$ critical

$$U(t) = A \cdot e^{-\alpha t} + B \cdot t \cdot e^{-\alpha t}$$

$$U(0) = A \overset{!}{=} U_0$$

$$\dot{U}(0) = -\alpha A + B \overset{!}{=} 0 \quad \Rightarrow \quad B = \alpha \cdot U_0$$

$$U(t) = U_0 \cdot e^{-\alpha t}\left(1 + \alpha \cdot t\right)$$